

SUPPLEMENTARY MATERIAL:

CLICKBAIT DETECTION

1. LSTM with AdamW Optimiser

To further evaluate the impact of optimisation strategies, the LSTM model was trained using the AdamW optimiser. The same model architecture and training configuration were used as in the Adam-based LSTM to ensure a fair comparison.

The AdamW model achieved the following performance:

- Accuracy: 0.9683
- Precision: 0.9780
- Recall: 0.9581
- F1-score: 0.9680

These results are comparable to those obtained using the Adam optimiser, with a slight decrease in overall accuracy. This supports the observation in the main report that additional regularisation through weight decay does not significantly improve performance for this dataset.

The confusion matrix for the AdamW-based LSTM model is shown below:

```
[[3131  69]
 [ 134 3066]]
```

The model demonstrates balanced performance across both classes, with a low number of misclassifications. The slightly higher number of false negatives compared to false positives indicates that the model is somewhat conservative when predicting clickbait. Thus, AdamW provides stable performance but does not offer a significant advantage over Adam for this task.

2. Additional Results: Word2Vec + SVM

The confusion matrix for the Word2Vec + SVM model is provided to further analyse its classification behaviour, as it was not included in the main report.

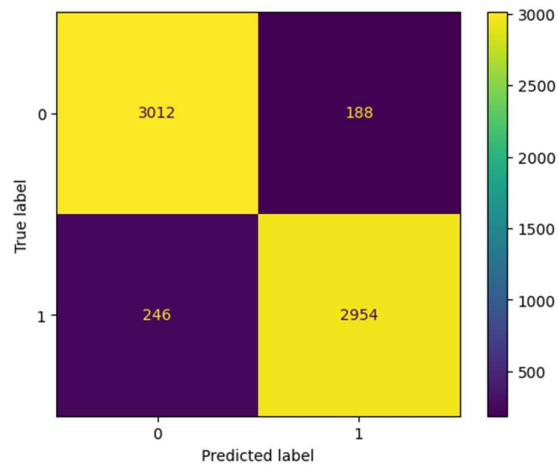


Figure: Word2Vec confusion matrix

The confusion matrix shows a higher number of misclassifications compared to the LSTM model. This reflects the limitation of the Word2Vec representation used in this study where word embeddings are averaged to form sentence-level features. As a result, word order and structural information are lost leading to reduced classification performance.

3. Hyperparameter and Training Setup

The following configurations were used to ensure consistency across experiments:

- **TF-IDF + SVM:**
max_features = 5000, ngram_range = (1,1), C (Regularization) = 0.1
- **Word2Vec + SVM:**
Word embeddings were generated for each token and averaged to form fixed-length sentence representations, which were then used as input to the SVM classifier.
- **LSTM Model:**
Tokenised and padded input sequences were used. The model consists of an embedding layer, an LSTM layer, and a dense output layer. The same architecture was used for both Adam and AdamW optimisers. Early stopping was applied to prevent overfitting.

All models were trained using the same train-test split to ensure comparability of results.

AI assistance declaration: I declare that the project objective, methodology design, experimental setup, and analysis presented in this project are my own work. I utilized AI tools to assist in refining technical explanations, improving report structure and clarity, debugging implementation issues, and polishing supplementary material. All final model implementation, preprocessing decisions, hyperparameter tuning, training, evaluation, and interpretation of results were performed independently.